STUDENT MARKS MANAGEMENT

Python Programming Exercise

Objective

Develop a Python program to manage students' marks in three subjects. The program should accept student details, calculate totals and averages, assign grades, identify the top-performing student, and generate a report that is saved to a text file.

Problem Statement

Write a program that allows the user to enter details and marks for multiple students. Each student has a name and marks in three subjects. The program must process the marks, display the results, and store the report in a file.

Requirements

- Ask the user how many students' details need to be entered.
- For each student, accept the student's name and marks in three subjects.
- Accept whole-number marks only. Each subject mark must be between 0 and 100.
- Calculate each student's total marks and average marks.
- Create a function to determine the grade based on the student's average:

Average Marks	Grade
90 and above	A
75 to below 90	B
60 to below 75	C
40 to below 60	D
Below 40	F

- Use exception handling to manage invalid numeric input.
- Display each student's name, subject marks, total, average, and grade.
- Identify and display the student with the highest total marks.
- Write the student results and the top student's details to a text file named student_report.txt.
- After saving the report, open the file and display its contents.

Questions to Consider

- How will you store the details of each student while processing the data?
- How will you ensure that marks entered are between 0 and 100?
- Why is it useful to create a separate function for calculating the grade?
- How will you identify the student who has scored the highest total marks?
- Which file mode should be used to create and write the report?
- How will you read the saved report and display its contents?

Expected Output

The program should display a clearly formatted summary of every student's marks, total, average, and grade. It should also display the top-performing student and save the same report in student_report.txt.